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HERITAGE RAILWAY ASSOCIATION

GUIDANCE NOTE

MATERIALS & Non-Destructive Testing

Supporting the Steam Locomotive Boiler series of Guidance Notes

Purpose

This document describes good practice in relation to its subject to be followed by Heritage Railways, Tramways and similar bodies to whom this document applies.

Endorsement

This document has been developed with, and is fully endorsed by, Her Majesty's Railway Inspectorate (HMRI), a directorate of the Office of Rail Regulation (ORR).

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1. Introduction

This Guidance Note is one of a series dealing with locomotive boilers that were produced by the 2006-8 meetings on “Steam Locomotive Boiler Codes of Practice”.

Railway locomotive boilers are designed to create, store and distribute steam at high pressure. The working life of such a boiler can be considerably shortened if due care is not taken at all stages of inspection, repair, running maintenance and day-to-day running.

In the past there have been a series of accidents and explosions owing to work being undertaken without having due regard to the inherent risks involved. It is with that in mind that HMRI and HRA set up the series of meetings of boiler practitioners to discuss the issues, distil good practice and codify it into this series of Guidance Notes.

This guidance is written for the assistance of persons competent to perform these tasks. In places the terminology used may be specific to such practitioners.

This guidance should also be useful to those in a supervisory or more general role. However no work should be undertaken unless the persons concerned are deemed competent to do so.

Where managements decide to take actions that are not in agreement with these recommendations, following appropriate risk assessments or for other reasons, it is recommended that those decisions are reviewed by the senior management body of the organisation concerned and a formal minute is recorded of both the reasons for and the decision reached.

2. ## MATERIAL PROPERTIES

Many people involved in boiler preservation and maintenance will come from walks of life which are not associated with engineering and may thus have little or no knowledge of the strength of the materials that they work with or of their physical properties. Often, corroded or wasted materials have to be renewed and the selection of the most appropriate equivalent materials from the range currently stocked by suppliers must be carefully made. This Section is included to illustrate the basic principles of strength of materials and their physical properties as related to the most common steels and non-ferrous metals encountered in boiler maintenance and as an aid to selection of appropriate materials. It is not intended to facilitate the design of modifications and readers must appreciate that the information included herewith can only be for initial guidance. The subject is vast and where knowledge beyond the content of this Section is required, the reader should seek further information from credible sources.

3. Stress and Strain

When a material is subjected to external forces, internal forces are set up which resist them. These internal forces are referred to as stresses.

Stress is defined as force per unit area or expressed algebraically as: $$Stress\ (\sigma) = \frac{Force}{Area}$$

Force is measured in:

- Pounds [lb]
- newtons [N]

Stress is measured in:

- pounds per square inch or
- lb/square in
- psi.
- newtons /sq.mm (Mega Pascals [MPa])

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Area is measured in:

- square inches
- sq.mm

There are three fundamental types of stress:

- Tensile stress
- Compressive stress
- Shear stress

Other derived stress conditions include “bending,” “torsion,” “fatigue” and “creep”

Manipulation of dimensions:

To convert N/sq.mm to psi – multiply by 145.037

To convert N/sq.mm to Ton/sq. in – multiply by 0.0647

1 N/sq.m = 1Pa (Pascal)

1 psi = 6895 Pa

1 M Pa = 1 N/sq.mm = 145 psi

1 lbf = 4.448 N

1 Tonf = 9.964 kN

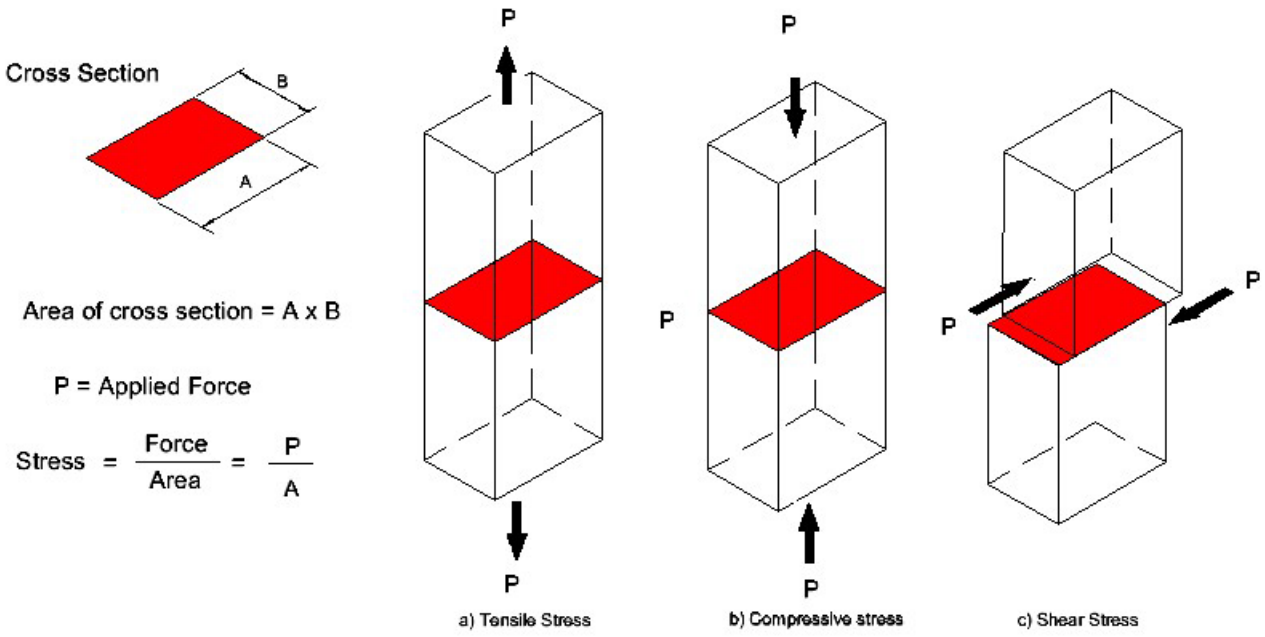


Fig 1 Fundamental Stress Applications

Fig. 1a) shows a metal bar of uniform cross section under the influence of a tensile load P which is spread over the area normal to it. Fig. 1b) shows the same metal bar under the influence of a compressive load. If the loads are applied as shown in Fig. 1c) parallel to each other and across the bar, the resulting stress is said to act in shear.

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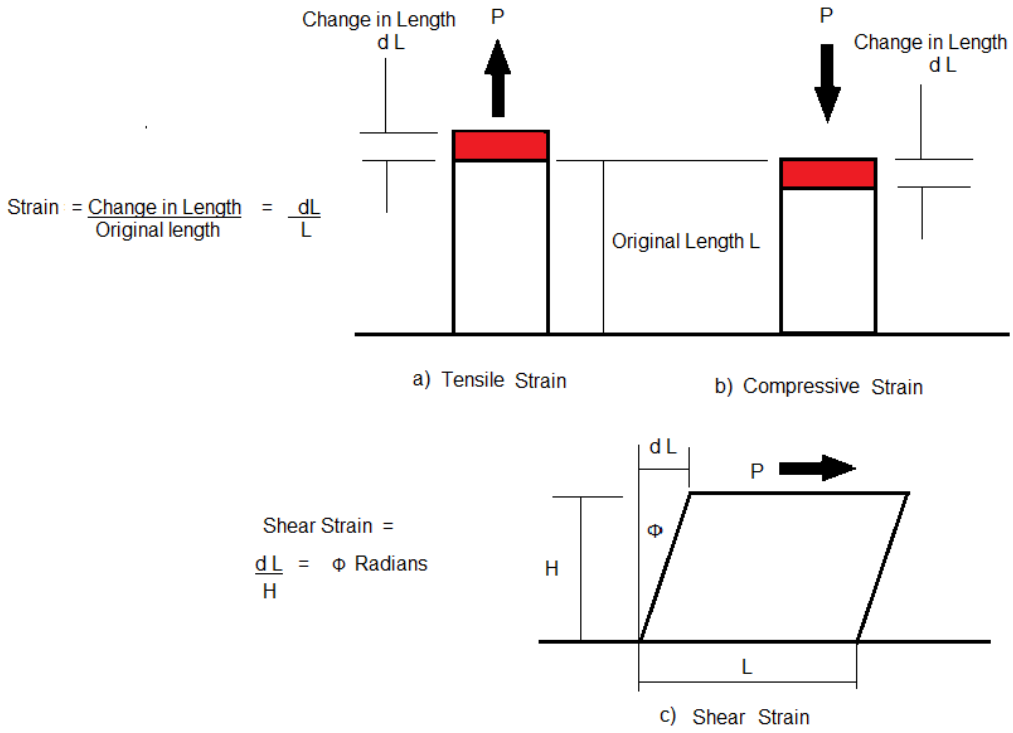


Fig 2 Fundamental Strain Applications

Strain is a measure of the change in shape of a material under the influence of stress. For the bar of metal shown in Fig. 2, the areas shown in red illustrate a) tensile, b) compressive and c) shear strains.

Strain is defined as change in dimension compared with its original dimension under the influence of stress, or expressed algebraically as:

$$Strain\ (\varepsilon) = \frac{Change\ in\ dimension}{Original\ dimension}$$

Since the components of strain are both lengths, the units of strain are expressed as a decimal or percentage.

It can be appreciated that a boiler, which is loaded under the influence of steam pressure, will have stresses set up in its structure which will give rise to corresponding strains. Strains must not be confused with expansion which is an entirely different mechanism and is due to the influence of temperature causing changes in dimension.

It will be useful to consider some recognised definitions of a material's particular key characteristics:

- Elasticity – this is the property that allows a material to change its shape under the influence of a load and return to its original size once the load has been removed.
- Plasticity – A material becomes plastic when subjected to sufficiently high stresses either in tension or compression such that, when the stresses are removed, the resulting deformation is non-reversible.
- Ductility – This refers to a property which allows a solid material the ability to deform under tensile stress, for example, by the material's ability to be stretched into a wire.
- Malleability – is a similar property to ductility and relates to a material's ability to deform under compressive stress, for example, by the material's ability to form a thin sheet by hammering or rolling.

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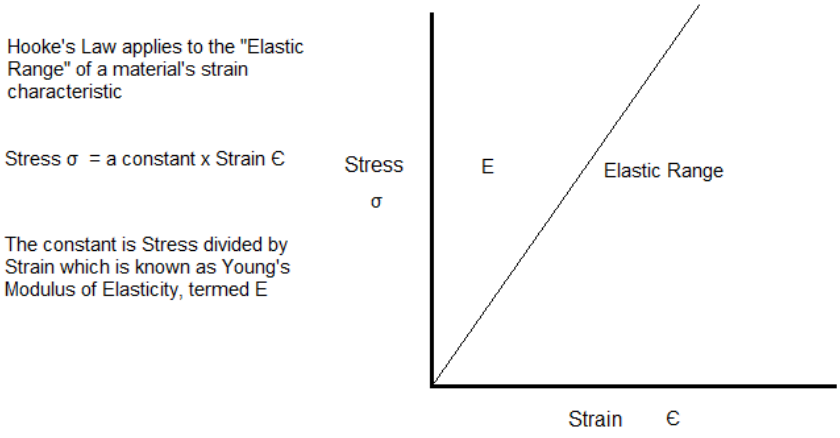


Fig 3 Hooke's Law

Stresses and strains are intimately related and over a material's elastic range of strength, this is according to Hooke's Law which is illustrated in Fig. 3. Hooke's Law states that within the elastic range the strain produced is directly proportional to the applied stress.

Most metals, and particularly the steels, have a defined range over which they are elastic. Many materials, however, do not have such a pronounced linear portion of the stress – strain characteristic but still return to their original size when stress is removed. There is a point, however, whereby continuing to apply stress starts to rapidly increase the strain and the material ceases to be elastic and starts to deform plastically. This point is termed the yield point.

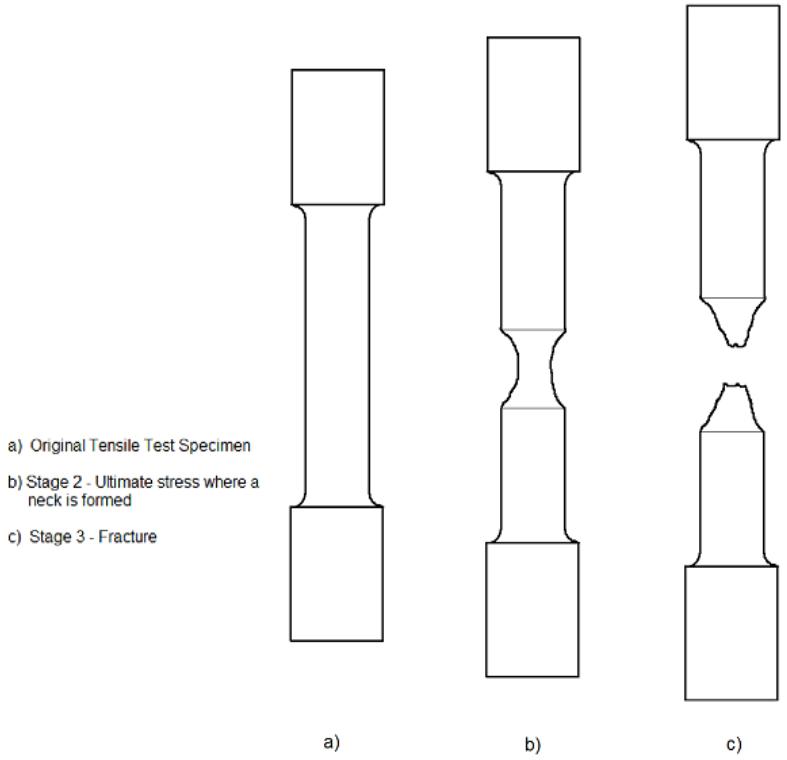


Fig 4 Stages of an axially applied tensile test on a mild steel specimen

When materials are tested, a specimen is machined out of a sample of the wrought material or from a feature of a manufactured item where specific values of strength are required to be established. Fig. 4 depicts three stages of progression during an axially applied tensile test on a mild steel specimen. As the load increases, the specimen with overall length shown at a) increases in length and after reaching the yield point, a neck starts to form as shown at b). After passing the point of maximum load, the neck area reduces rapidly and since the load is spread over a diminishing area, the stress increases. Finally, the specimen fractures as shown at c).

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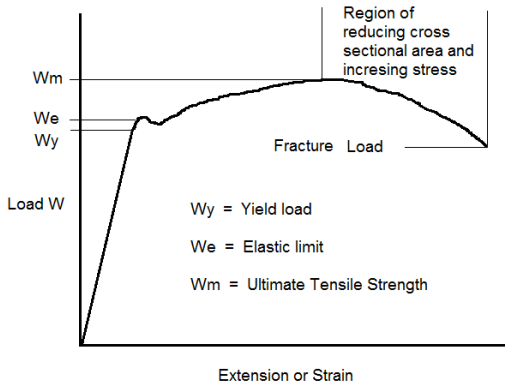


Fig 5 Nominal Stress Strain Diagram - typically for mild steel

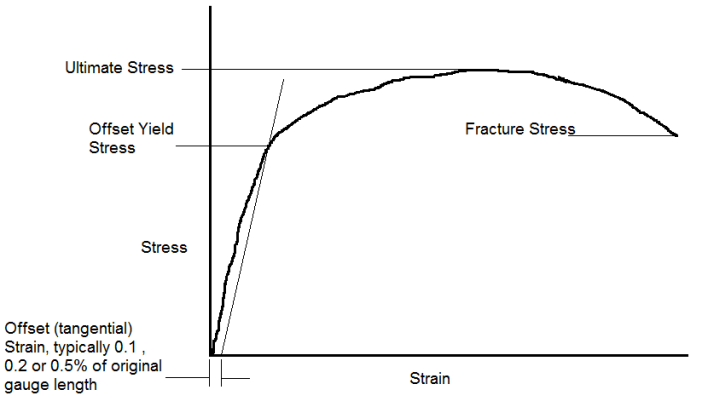


Fig 6 Stress - Strain diagram for material with a non linear strain characteristic

In Figs. 5 and 6, the relationship between stress and strain is shown for mild steel and more ductile materials such as copper respectively. They are both ductile materials and Fig. 5 is termed a Nominal Stress – Strain diagram. In this diagram, the value of stress is determined by dividing the load applied to the specimen by its original cross sectional area. The linear portion of the characteristic portrays the elastic range up to point Wy. At point We the elastic limit is reached, following which, the application of further stress initiates the formation of a neck. Point Wm represents the maximum load which can be carried by the specimen which is termed the Ultimate Tensile Strength or UTS. At this point the load decreases rapidly as the cross section reduces and the stress actually increases to the point where the specimen fractures.

The algebraic expression of Ultimate Tensile Strength is:

$$Ultimate\ Tensile\ Strength = \frac{Maximum\ load}{Original\ area\ of\ section}$$

Copper and other ductile materials do not exhibit a definite elastic range and it is not possible to nominate a definitive yield point. Referring to Fig. 6, it will be seen that the characteristic stress – strain relationship follows a curve without a linear portion. The Yield Point is thus not obvious and is usually defined as a percentage off set where a definite or stated amount of permanent set, typically 0.1%, 0.2% or 0.5% of the original gauge length takes place. The UTS is at the highest point on the curve which as can be seen extends to allow a great deal of strain before fracture takes place. This characteristic is representative of a ductile material. For a hard material, very seldom if ever used in a boiler, the yield point is determined in a similar manner and is termed the proof stress, typically associated with a strain of 0.1% or less.

Engineers use another derived constant known as Young's Modulus of Elasticity which is used to determine the stresses set up in a material that exhibit a clearly defined linear portion of the stress – strain characteristic. The value of this constant is the slope of the line and is expressed algebraically as:

$$Youngs\ Modulus\ of\ Elasticity\ (E) = \frac{Stress}{Strain}$$

In working with materials subjected to shearing forces, engineers use the term Modulus of Rigidity. The value of this constant is expressed algebraically as:

$$Modulus\ of\ Rigidity\ (G) = \frac{Shear\ Stress}{Shear\ Strain}$$

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4. Examples of Fundamental Stress Calculations

Hoop and Longitudinal Stress

Perhaps the most fundamental stresses set up in a cylindrical pressure vessel are the hoop and longitudinal stresses. The following equations are only valid for thin wall cylindrical pressure vessels but are appropriate for the normal thicknesses used in boilers.

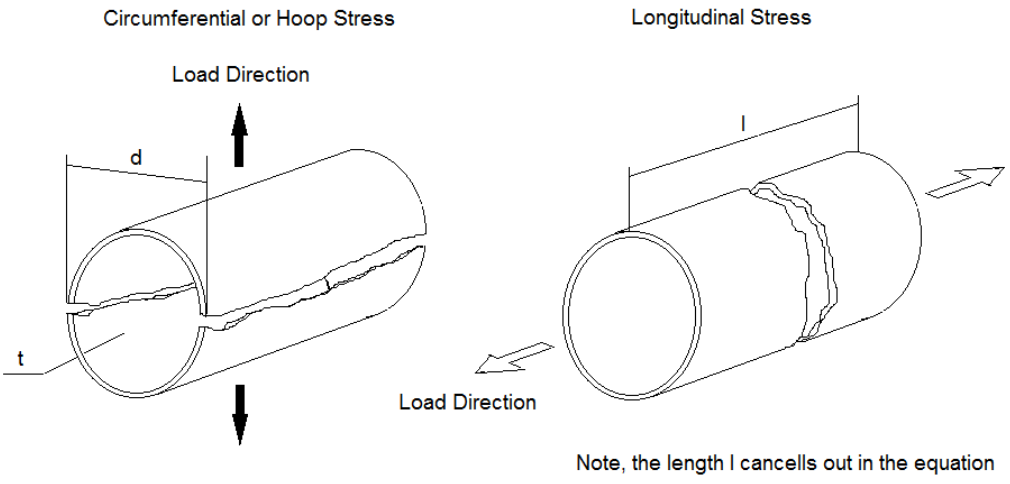


Fig 7 Stresses in cylindrical thin wall cylinders

In the shell illustrated in Fig. 7, the following apply:

p = internal pressure

d = outside diameter

l = length of cylinder

t = thickness of plate

σ_1 = circumferential or hoop stress

σ_2 = longitudinal stress

The circumferential or hoop stress equates to: $$\sigma_1 = \frac{pd}{2t}$$

The longitudinal stress equates to: $$\sigma_2 = \frac{pd}{4t}$$

At once it can be seen that the forces tending to tear the shell along its length are double those tending to force it apart around its circumference. For this reason it will be evident that the methods used to join longitudinal seams have to be stronger than those for circumferential joints.

Factor of Safety

A designer or stress engineer is responsible to ensure that the design of the work not only fulfils its intended purpose but is also safe to use. This introduces the concept of safe working stress. In selecting the material to be used, the UTS will be established from the material specification. It is never acceptable to allow the working stress to approach the value of UTS and all designers of structural parts will only allow the working stress to be limited to a fraction of this amount and very often to a fraction of the yield point or off set stress. This fraction is referred to as the factor of safety. This is expressed algebraically as:

$$Factor\ of\ Safety = \frac{Ultimate\ Stress}{Safe\ Working\ Stress}$$

For example, a material having a UTS of 25 tons per sq. in. may be assigned a safe working stress of 5 tons per sq. in. giving a Factor of Safety of 5.

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When considering shear loads, Von Mises criterion is employed which demonstrates that the yield stress in pure shear (σ_k) is proportional to the tensile yield strength of the material (σ_y) divided by √3 or expressed algebraically as:

$$\sigma_k = \frac{\sigma_y}{\sqrt{3}} \qquad \text{or may alternatively be expressesd as } 0.577\sigma_y$$

Riveted Joints

Most heritage boilers are of riveted construction. There are several types of riveted joint, the most common being the lap type where one plate overlaps the other as shown in fig. 8a). Examples of this type of joint are the circumferential seams of boiler barrel rings, tubeplates and for firebox flanged plates. In all these examples, the rivets are stressed in single shear. Doubling plates are used for the longitudinal joints where the rivets are stressed in double shear as illustrated in Fig. 8b). It should be noted that in early boilers, longitudinal barrel joints were usually lap seamed and due to the combined effect of pressure and corrosive conditions, the plates had a tendency to groove and many catastrophic failures resulted as a consequence.

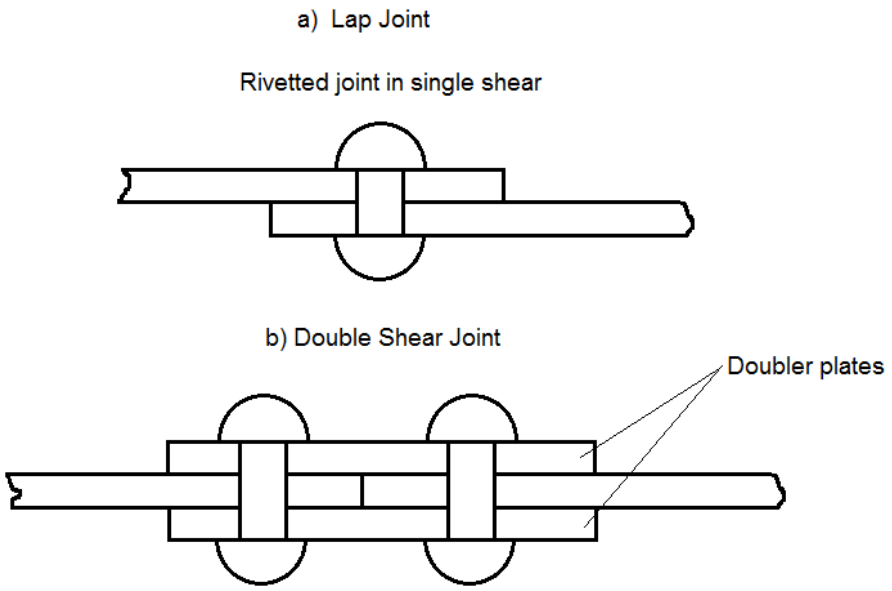


Fig 8 Typical Rivetted joints used in boiler construction

The thickness of the plates is defined by the loads they have to carry. Once this is established (see below) the diameter and pitch of the rivets can be calculated.

It will be apparent that the strength of a riveted joint will never reach the same as that of the parent plate. The concept of joint efficiency is introduced to deal with this fact as follows:

$$Efficiency\ of\ \ Joint\ = \frac{Strength\ of\ \ Joint}{Strength\ of\ \ original\ \ plate}$$

The following value of ultimate stresses is typical of those used in boiler design:

		Ultimate Stress (Tons per square in)
Tension	Steel	28
	Copper*	14
Shear	Steel	23
	Copper*	9.5
Compression	Steel	46
	Copper*	16

*Arsenical grade copper

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Riveted joints usually use a factor of safety of between 5 to 6 and the following joint efficiencies:

- Single-riveted joint - 50 to 55%
- Double-riveted joint - 65 to 70%
- Treble-riveted joint - 80 to 85%

Using the example of the pressure vessel, the means of determining the plate thickness is to use the circumferential stress as previously calculated. As an example, if the Factor of Safety is 5 and the Joint efficiency, η is 70% the safe working stress would be calculated as follows:

$$Safe\,Working\ Stress = \frac{UTS\ of\ the\ plate}{Factor\ of\ Safety}$$

Assuming the previous example, the Safe Working Stress would be 5 tons per sq. in.

The theoretical thickness is then calculated from the formula for Circumferential or Hoop stress. The required thickness is then established as:

$$Required\ \ thickness = \frac{Theoretical\ thickness}{Efficiency\ of\ the\ joint\ (0.7)}$$

Having determined the Required Thickness, there are two more considerations in the design of a boiler to take into account before deciding the final thickness of the plates. These are corrosion and temperature. Normally a corrosion or temperature factor is applied where required. To illustrate this, if a corrosion factor of 0.8 is used and a temperature factor of 0.95, then if the Required Thickness is 0.5" the final thickness would be:

$$Final\ \ thickness = \frac{0.5}{0.8\ x\ 0.95} = 0.657''$$

The nearest equivalent Imperial plate thickness would be 0.656 or 21/32".

An empirical formula to determine the diameter of the rivets is:

$$Diameter\ of\ rivet\ \ = \ \ 1.2\sqrt{Plate\ thickness}$$

The pitch is the uniform distance between the centres of the rivets and although a line of rivets may extend over a particular length, it is only necessary to consider the conditions in a length of joint equal to the pitch of the rivets.

The following basic facts relate to the ways riveted joints can fail:

- The rivet may shear;
- The plate may tear between the rivets; or
- The plate or rivet may crush.

The resistance to shearing of a rivet is the cross sectional area of the rivet of diameter d multiplied by its shear strength.

$$Shear\ strength\ of\ rivet\ \ = \ \ \frac{\pi d^2\ \sigma_{shear}}{4}$$

The resistance to tearing of the plate is the area of the plate between the inside of the adjacent rivets or expressed algebraically as:

$$Tearing\ Shear\ strength\ of\ \ joint\ \ = \ \ \left(pitch\ of\ \ rivets - diameter\ of\ \ one\ rivet\right) thickness\ of\ \ plate\ x\ tensile\ stress$$

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Whilst the crushing strength of a riveted joint is not normally an issue, it is usually calculated as a check to ensure it is below the primary stresses. This is achieved by multiplying the projected area of a rivet through one plate by the compressive strength of the plate.

To achieve the same strength of a riveted joint both in tension and in shear the following equation is used:

$$Pitch\ of\ rivets = \frac{\pi d^2\ \sigma_{shear}}{4\ x\ plate\ thickness\ x\ \sigma_{tensile}} + d$$

Although British Standard B.S. 931 for Loco-Type Boilers of Riveted Construction is now obsolete, for those that are concerned, the rules for riveted joints are set out in its contents. These include single, double and other configurations of riveted joints.

5. Other Related Stress Factors

Metal Fatigue

The stress patterns within a boiler's structure are very complex. It is a tribute to the design of the locomotive boiler that so many hundreds of thousands have provided a safe means of generating steam during the span of the age of steam traction. In the main, designers of boilers very often employed rule of thumb principles with a few first order calculations to bolster the assumptions. The main problem was (and still is) that what goes on inside a boiler is largely a matter of conjecture. For example, the complex stresses in boiler stays are difficult to ascertain and many attempts at solutions to premature fracturing have been tried. There is no doubt that many failures of boiler components and structures result from thermal cycling. Such cycling is due to the strains produced when lighting up from cold and cooling down from hot and more critically from heavy fluctuations of output during operations in a railway environment. Some appreciation of what this entails may be found useful.

The cause of premature fracture of materials under fatigue or cyclic stress conditions was not fully understood until the coming of the aeronautical engineering industry. Items of engine mountings, undercarriages and wings failed without warning, a truly disastrous situation. This led to intensive investigation of the phenomenon with the result that in present times the onset of fatigue failure can be readily predicted and actions instituted to prevent catastrophic failures. In boilers, the principles on which the fatigue theory is based also apply in certain areas of the structure.

Considering the mechanism of cyclic stresses: The stresses involved can be applied mechanically or indirectly by thermal means and failures are influenced both by the level of stress and to the number of cycles to which the particular structural element is subjected.

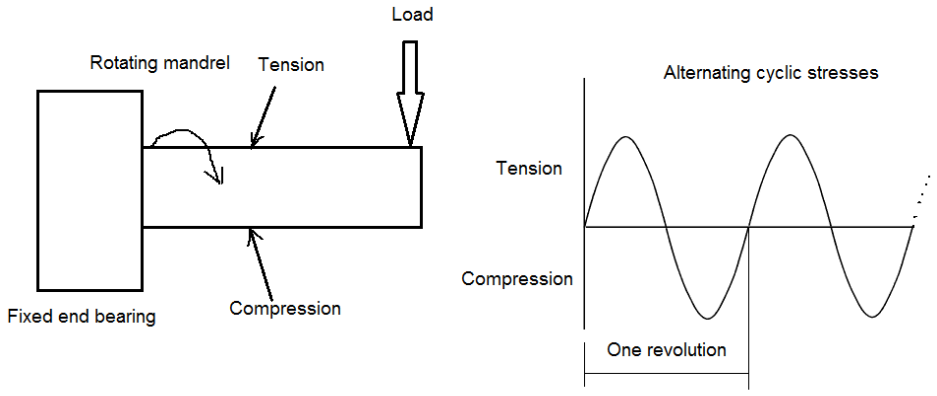
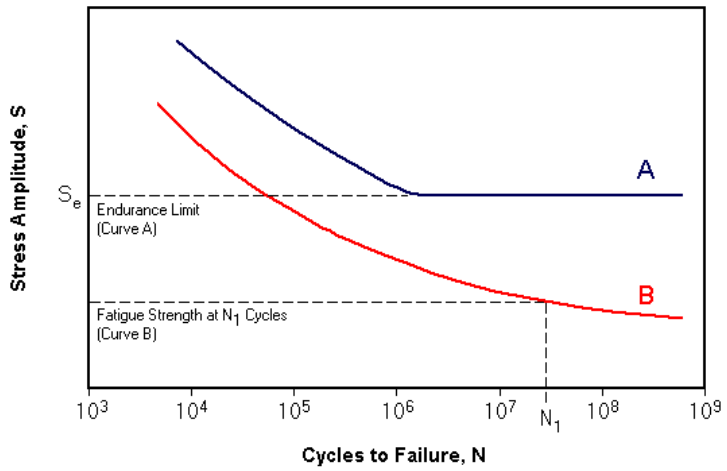


Fig 9 Rotating cantilever subject to alternating stresses

Fig. 9 shows a rotating spindle with load acting on its free end, this load being reacted by roller bearings at both ends. It will be apparent that the fibres in the top of the spindle are in tension, whilst those at the bottom are in compression. As the spindle rotates through one revolution, the stresses alternate from tension to

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compression and during continued cycles, the spindle becomes subject to fatigue. This is a similar effect to bending a piece of thin metal to and fro. Eventually the specimen breaks.



Curve A denotes a material that has a fatigue limit. A fatigue limit (also called an endurance limit) denotes the critical stress below which fatigue fracture will not happen. A material stressed to the endurance limit is said to have "infinite life."

Curve B denotes a fatigue curve that does not have a fatigue limit. It does however have fatigue strengths. That is to say for a given stress thereis a given fatigue life or number of cycles to failure of that material. The maximum number of cycles it takes for a material to fracture is called fatigue strength.

Fig 10 Example of Stress / Cycles curves

Testing metals for fatigue resistance is now a common practice and the characteristic σ/n curve is shown in Fig. 10. The curve represents the varying points at which a specimen will fracture due to the cumulative effect of cyclic stress. If the stresses are very high, the onset of fracture is relatively quick. However at the other end of the curve, provided the stress applied is below the curve which eventually levels out, the material will not fracture and is said to have infinite life. Where boilers are concerned, it is very difficult to predict the cycles to failure since the applied stresses are mainly thermally generated and also under the influence of other factors.

This, unfortunately, is not the whole story, since materials which are subject to fatigue loading are usually very sensitive to the effect of surface finish and particularly so to surface breaking defects such as indentations, abrasions or flaws resulting from the manufacturing process. The rules for inspection of axles, for example, preclude their use if indentations caused by ballast or other factors exceed a certain size or are present in areas of high stress. Not so common in present days, due to the fact that modern steels produced in the electric arc furnace are much more pure than with older processes, is the formation of internal faults where the effect of fatigue tends to propagate a crack from within, such as was alarmingly prevalent in running rails in the days of open hearth steel production. Once a surface defect is present, the probability of crack initiation becomes very high in a fatigue environment and, eventually, a crack so caused, propagates to failure. Some alloy steels have low notch sensitivity and others are designed to have a very high resistance to such faults. Alloy steels are not normally used in boiler construction although the use of higher tensile steels containing a small quantity of Nickel has been employed, notably by the LMSR, aimed at reducing the weight contribution of boilers where rail or bridge loading issues were prevalent.

To sum up, some areas of a boiler tend to suffer from the effects of metal fatigue, most prevalently staybolts and particularly in the breakage zones. The effect of alternating thermal inputs also causes strains and their associated stresses. Staybolts, tube plates, door plates, back plates and throatplates are all affected by this mechanism and are evidenced by the formation of cracks which if left unchecked, will eventually result in failure.

Stress Corrosion Cracking

Since a boiler operates, by virtue of its very function, with a potentially corrosive filling, unless treated, the boiler water will aid the process of corrosion. Add a stress component to the corrosive environment and conditions become conducive to stress corrosion cracking. The mechanism of this failure mode is particularly prevalent on plates that are subject to local intense stresses such as adjacent to the foundation ring. Here, the effect of corrosion and local high bending stresses gives rise to what is commonly termed grooving. Once this form of corrosion reaches a critical stage, the corrosive action initiates a crack which eventually propagates through the plate. Effective water treatment will normally eliminate this problem.

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Creep

When a material is subjected to high stress at high temperatures, it can extend or change its physical shape. During such time, the material becomes weakened and can fail to perform its intended role. Such temperatures and conditions do not occur in locomotive boilers although can become troublesome in high performance watertube boilers used in power generation.

6. Physical Properties of Boiler Steels

The chemical composition of boiler steels is given in other parts of these Codes of Practice as are the tensile and yield limits. In general terms, steels for boilers must essentially be of a ductile nature. The properties of all steels are highly influenced by the amount of carbon that they contain. To achieve a ductile or energy absorbing condition, the percentage of carbon should not generally exceed 0.13%. A measure of the ductility of steel plates used in boiler construction or repair is the result of an impact “Charpy” or “Izod” notch test. Fig. 11 refers.

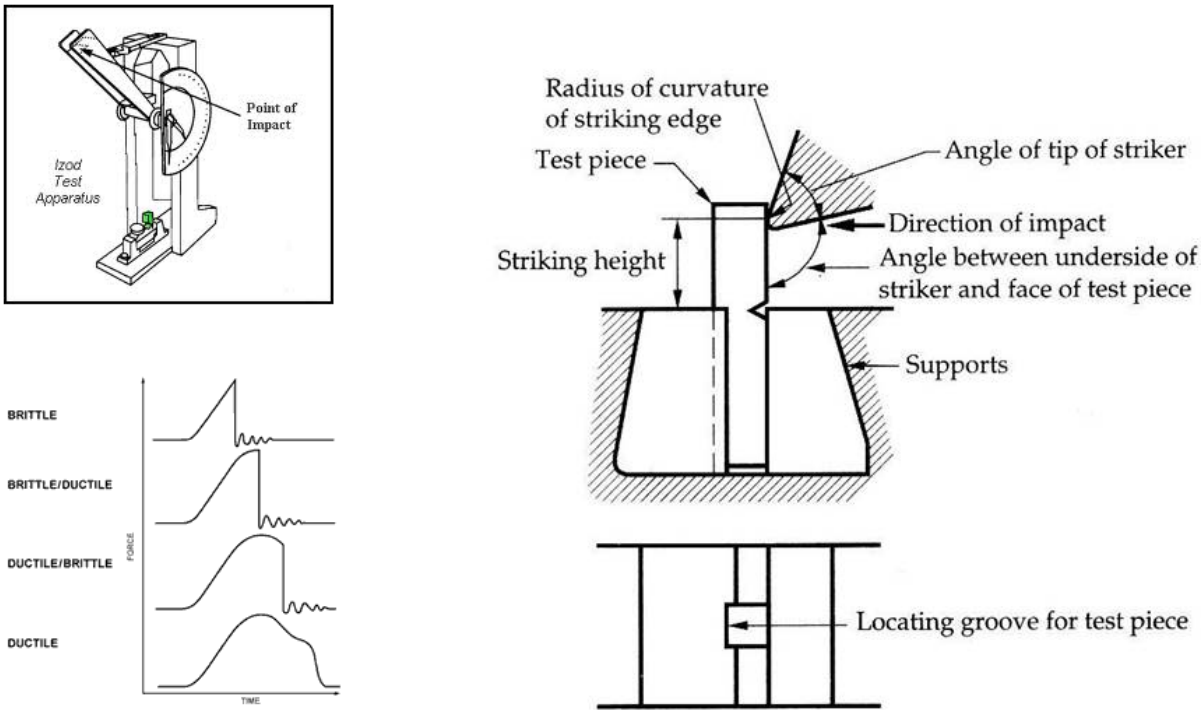


Fig 11 Typical Impact Testing Apparatus

A specimen of the material is produced and an impact applied in the testing machine to fracture it. Whilst the impact energy can be determined from the test apparatus, the results are subjective assessments and thus some caution should be exercised with the interpretation of energy absorption figures obtained. In cases of doubt, a professional interpretation of the fracture faces and local deformations may also be necessary when considering the suitability of a material.

One test that can readily identify if a material is ductile is to prepare a sample strip of material and bend it back on itself through 180° over a 1 x its thickness radius bend. If the metal “tears” on the outside of the bend, it should be regarded as suspect and more exhaustive tests carried out.

Another property that can readily be assessed is hardness. It is often useful to know the hardness of a material either to verify if it is representative of the material in question or if local hardening has taken place which may lead to subsequent failure. This can be carried out on a specimen of the parent material. Several hardness testing methods are available, the more common ones being the Vickers, Rockwell and Brinell.

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These all involve indenting the surface under examination using a hardened element and measuring the size of the indentation.

Where “high tensile” steel boilers are encountered, it is safer not to undertake weld repairs. However, if such repairs prove necessary, it is essential to ensure that the welding process is carried out in accordance with a credible Welding Procedure which must be approved by the competent person (as is also the case with other welds on boiler structures).

7. Associated material terminologies

The following descriptions outline some of the terms used to describe or comprise properties of various metals:

Alloy: A substance having metallic properties and composed of two or more chemical elements of which at least one is metal. Usually possesses qualities different from those of the components.

Alloy Steel: Steel containing significant quantities of alloying elements other than carbon and the commonly accepted amounts of manganese, silicon, sulphur, and phosphorus.

Annealing: Heating to and holding at a suitable temperature, followed by cooling at a suitable rate to lower the hardness or alter other mechanical or physical properties.

Carbide: A compound of carbon with one or more metallic elements.

Carbon: Element occurring as diamond and as graphite. Carbon reduces many metals from their oxides when heated with the latter, and small amounts of it greatly affect the properties of iron. Though classed as non-metallic, metallurgically, like boron, it is treated as a metal.

Cold Work: Plastic deformation of a metal at room temperature. Substantial increases in strength and hardness may occur.

Cold-shortness: Large quantities of phosphorus (in excess of 0.12%P) which reduces the ductility, thereby increasing the tendency of the steel to crack when cold worked. This brittle condition at temperatures below the recrystallization temperature is called cold-shortness.

Cooling Curve: A curve showing the relationship between time and temperature during the solidification and cooling of a metal sample. Since most phase changes involve evolution or absorption of heat, there may be abrupt changes in the slope of the curve.

Corrosion: 1) Gradual chemical or electrochemical attack on a metal by atmosphere, moisture or other agents, and 2) chemical attack of furnace linings by gases, slags, ashes or other fluxes occurring in various melting practices.

Crack, Hot Tear: A rupture occurring in a casting or welded joint at or just below the solidifying temperature by a pulling apart of the soft metal, caused by thermal contraction stresses.

Crystalline Fracture: Breakage of a brittle metal, showing definite crystal faces in the fractured surface.

Defect: A discontinuity in the product whose severity is judged unacceptable in accordance with the applicable product specification.

Elongation: The percentage increase in length of a part under tensile test just before it fractures.

Embrittlement: Loss of ductility of a metal due to chemical or physical change.

Ferrite: A solid solution of one or more elements in the body-centre-cubic phase of iron or steel.

Ferritic Steels: Steels in which ferrite is the predominant phase. These steels are magnetic.

Hot Shortness: Hot shortness means that the steel will literally break apart during hot rolling or working resulting in failure. Sulphur in steel is considered injurious except when added to enhance machinability. Sulphur readily combines with iron to form a low melting iron sulphide. Sulphur causes hot shortness in steel unless sufficient manganese is added. Sulphur has a greater affinity for manganese than iron and forms manganese sulphide which has a melting point above the hot rolling temperature of steel, which eliminates hot shortness.

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Hot Tear: A crack or fracture formed prior to completion of metal solidification as a result of hindered contraction. A hot tear is frequently open to the surface of the casting and is commonly associated with design limitations.

Hydrogen Embrittlement: A condition of low ductility resulting from the absorption of hydrogen, often caused by welding in the presence of moisture. A time dependent fracture process which results in a loss of ductility.

Impurity: An element unintentionally allowed in a metal or alloy. Some impurities have little effect on properties; others will grossly damage the alloy.

Inclusions: Non-metallic materials in a metal matrix. Sources include reoxidation, refractories, slag, and deoxidisation products.

Ingot: A mass of metal cast to a convenient size and shape for remelting or hot working.

Intergranular Corrosion: Corrosion in a metal taking place preferentially along the grain boundaries.

Iron: 1) A metallic element, melting point 1535°C (2795°F), 2) Irons not falling into the steel categories, as Gray Iron, Ductile Iron, Malleable Iron, White Iron, Ingot, and Wrought Iron.

Lateral Expansion: A measured property used in Charpy Impact Testing. This refers to an increase in width of a specimen after fracture.

Dye Penetrant Testing: A non-destructive testing method suitable for evaluating the surface integrity of non-magnetic and ferro-magnetic parts. (see section on NON-DESTRUCTIVE TESTING)

Magnetic Particle Inspection: A non-destructive method of inspecting the surface integrity of ferromagnetic materials. (see section on NON-DESTRUCTIVE TESTING)

Malleable Iron: A mixture of iron and carbon, including smaller amounts of silicon, manganese, phosphorus, and sulphur, which after being cast (white iron, carbon in combined form as carbides) is converted structurally by heat treatment into a matrix of ferrite containing nodules of temper carbon (graphite).

Metallographic Structure: The nature, distribution, and amounts of the metallographic constituents in a metal.

Mechanical Properties: Properties of a material that reveal its strength and elastic behaviour.

Metallurgy: The science and technology of metals, a broad field that includes, but is not limited to, the study of internal structures and properties of metals and the effects on them of various processing methods.

Microspectroscopy: A method of identifying metallic constituents using spectrographic arc.

Mild Steel: Plain carbon steel of about 0.25% carbon or less.

Mill Scale: Iron oxide scale formed on steel during hot working processes, cooled in air.

Non-destructive Testing (Inspection): Testing or inspection that does not destroy the object being tested or inspected. (See section on NON-DESTRUCTIVE TESTING.)

Non-ferrous: An alloy in which the predominant metal or solvent is not iron.

Normalising: Heating a ferrous alloy to a suitable temperature above the transformation temperature Ac3, followed by cooling at a suitable rate, usually in still air to a temperature substantially below the transformation range.

Oxidation: Any reaction of an element with oxygen. In a narrow sense, oxidation means the taking on of oxygen by an element or compound.

Oxide: A compound of oxygen with another element.

Pitting: A form of wear characterised by the presence of surface cavities, the formation of which is attributed to processes such as fatigue, local adhesion, cavitation or corrosion.

Physical Properties: The various attributes of a material including its strength, ductility, hardness, density, coefficient of expansion, etc.

Plastic Deformation: Permanent distortion of a material under the action of applied pressure.

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Residual Stress: Those stresses set up in a metal as a result of non-uniform plastic deformation, or the unequal cooling of a casting.

Scaling (Scale): Surface oxidation, partially adherent layers of corrosion products, left on metals by heating or casting in air or in other oxidizing atmospheres.

Spalling : Buckling or flaking-off of the surface material.

Stress-Corrosion Cracking: Spontaneous failure of metals by cracking under combined conditions of corrosion and stress, either residual or applied.

Stress Relieving: A heat treatment to reduce residual stresses, followed by sufficiently slow cooling to minimize development of new residual stresses.

Thermal Fatigue: Failure resulting from cycles of alternate heating and cooling, more critically caused by the difference in temperature between the hot face and cooler face generated in the heating cycle.

Thermal Shock: Stress developed by rapid and uneven heating of a material.

Toughness: The ability of the metal to absorb energy and to deform plastically during fracture. Toughness values obtained in testing depend upon the test temperature, the rate of loading, the size of the test specimen, as well as the presence of a notch and its acuity.

Transformation Temperature Ac3: The temperature at which transformation of ferrite into austenite is completed upon heating.

Ultrasonic Testing: A non-destructive method of testing metal for flaws based on the fact that ultrasonic waves are reflected and refracted at the boundaries of a solid medium. (see section on NON-DESTRUCTIVE TESTING)

8. Properties of Non-Ferrous Boiler Materials

The non-ferrous metals comprise brasses, bronzes, gun metals, copper and the cupro-nickel alloys.

The “yellow” metals are used for valves, casings, manifolds, pipe unions and a host of other boiler fittings where good resistance to corrosion is essential and mechanical friction is of a low order.

Traditionally, the LMR and BR non-ferrous metals were identified by the “YM” number or equivalent BR Specification number. These in turn relate to BS Specifications. Table 1 summarises these metals.

Monel Metal is often specified for firebox stays. It is usually supplied hot rolled to sufficiently accurate dimensions to allow screwing the ends without previous machining. Its leading physical properties are:

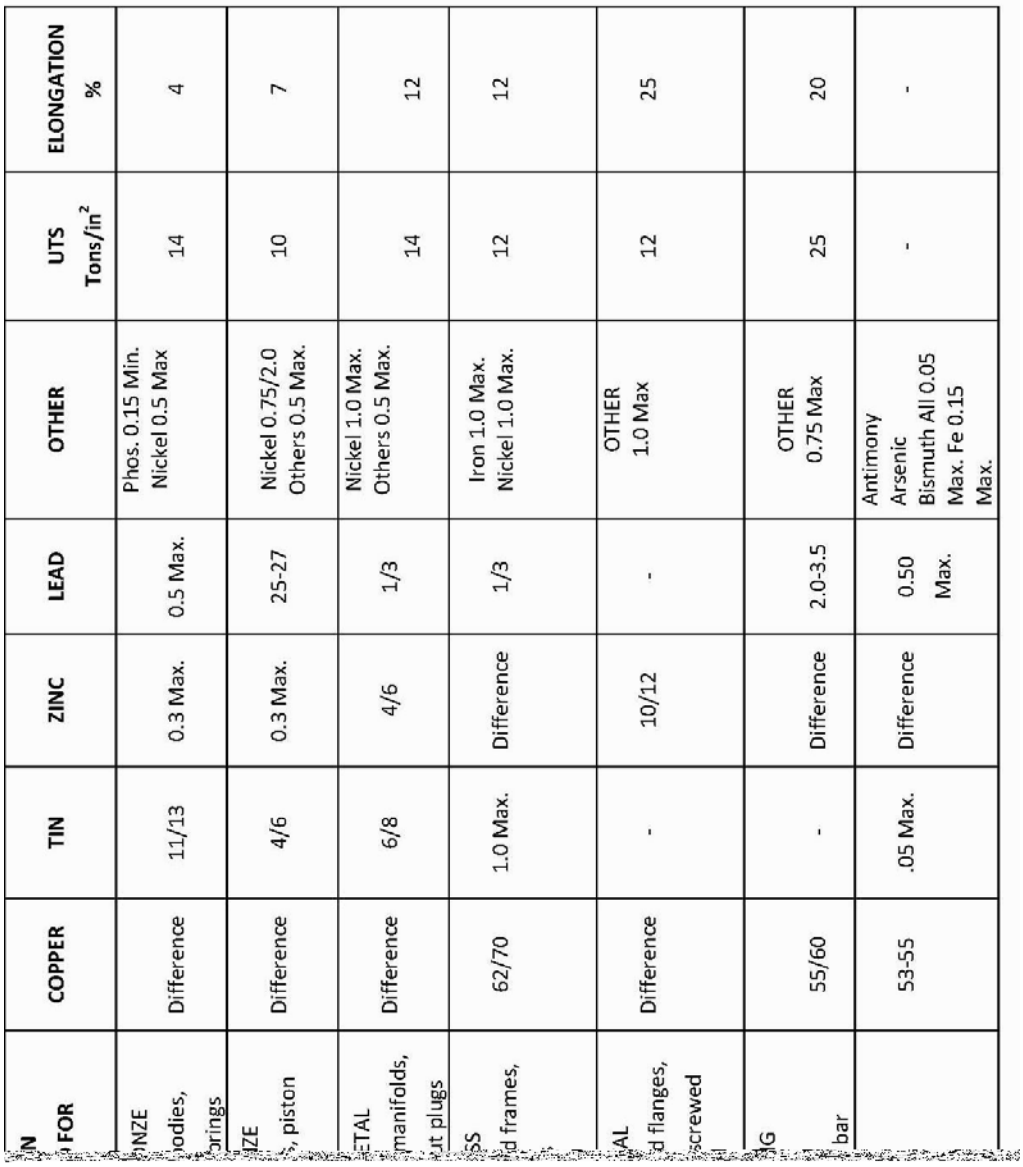
Ultimate Tensile Strength	32 - 38 tons/sq. in.
Yield Point	14 - 16 tons/sq. in,
Elongation	35%
Brinell Hardness	110 - 120

9. Purchasing New or Replacement Materials

When purchasing materials intended for renewal of worn or corroded boiler materials, the order should quote to the stock holders the National or International standards that the materials being purchased must conform to. The order should also state any special conditions to which the materials should be finished and the methods of inspection and delivery that are to be employed. A validated certificate of conformance must be obtained and held on record. The materials should be indelibly marked to verify traceability and the competent person should seek verification that suitable materials have been procured.

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10. Table 1: LMR / British Railways non-ferrous alloys



N FOR	COPPER	TIN	ZINC	LEAD	OTHER	UTS Tons/in^2	ELONGATION %
NZE odies, prings	Difference	11/13	0.3 Max.	0.5 Max.	Phos. 0.15 Min. Nickel 0.5 Max	14	4
ZE , piston	Difference	4/6	0.3 Max.	25-27	Nickel 0.75/2.0 Others 0.5 Max.	10	7
ETAL manifolds, ut plugs	Difference	6/8	4/6	1/3	Nickel 1.0 Max. Others 0.5 Max.	14	12
SS d frames,	62/70	1.0 Max.	Difference	1/3	Iron 1.0 Max. Nickel 1.0 Max.	12	12
AL d flanges, screwed	Difference	-	10/12	-	OTHER 1.0 Max	12	25
G bar	55/60	-	Difference	2.0-3.5	OTHER 0.75 Max	25	20
	53-55	.05 Max.	Difference	0.50 Max.	Antimony Arsenic Bismuth All 0.05 Max. Fe 0.15 Max.	-	-

Since this table was produced by British Railways; BS1400 has been replaced by BS EN 1982:2002 and BS249 has been replaced by ASTM B927/927M-09

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11. Table 2: Steel Equivalence Chart

The table below gives the modern specification equivalence for some common specification steels.

In order to prevent differing specification materials being mixed it is suggested that users determine their own colour codes.

Suppliers colour codes for steel types should not be used as the sole means of identification as not all suppliers use the same colours. The material certificate should always be checked.

BS970 1955 old EN	BS970 1970 modern
EN1 A	220M07 or 230M07
EN1 A leaded	220M07 pb
EN3	
EN3 A	070M20
EN3 B	080A15
EN3 C	070M20
EN5 K	080M30
EN8	080M40
EN16	605M36
EN 24	817M40
EN32	080M15
EN32 C	080A15

BS970 has now been split up into - BS EN 10095:1995, 10250-4:2000, 10085-2001,PD 970-2001, BS EN 10087:199, 10083-1-1991, 10084-1-1998 all might be applicable.

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12. ## NON-DESTRUCTIVE TESTING

The following NDT techniques are listed as the most commonly used and suitable techniques and thus the most often used techniques for the examination of various parts of Locomotive and related types of boilers and pressure vessels. The listed techniques are suitable for the examination of plate, welds and rivet metal joining techniques.

When selecting which type of technique is needed to confirm the integrity of the different parts to be examined care should be taken to ensure that the most suitable technique is employed.

The following criteria should be used when choosing the technique to be employed:

- What defects are being looked for?;
- What is the geometry of the area to be investigated?; and
- What material is to be examined?

When examining flat plate it is relatively easy to select the technique to be employed as most techniques can be used. It should however be noted that if radiography is to be employed access to both sides of the plate would be required.

Should access be limited it may not be possible to use an ultrasonic technique due to the restricted area for the operator to scan. Thickness of the material may also be a limiting factor particularly if thin for ultrasonic techniques. Another consideration would be a change of section of the material, which could restrict the amount of meaningful data that could be obtained from the area under test.

Basically NDT techniques are formed of two distinct groups:

- Surface breaking.
- Volumetric – which is to say what is contained within a piece of material, often weld material, but could be defects contained within non-worked plate.

Surface Breaking Techniques

These techniques are commonly Magnetic Particle or Dye Penetrant Inspection and are solely concerned with the location of surface breaking defects such as cracks between rivet holes. These techniques give no indication of defects within welds although Magnetic Particle Inspection can pick up sub-surface defects.

Magnetic particle inspection is the preferred technique for location of surface breaking or near surface defects when examining areas on magnetic (ferrous) materials. Where the material being tested is non-magnetic other techniques such as dye penetrant testing should be employed.

Volumetric Techniques

These techniques are commonly Ultrasonic or Radiographic Examination. Both techniques, when properly employed, give excellent results. The employment of these two techniques very much depends on the geometry of the area to be tested. With ultrasonic techniques a clear area to allow complete scanning is required, whereas, with radiography, changes of material section and general access to the test area may be a consideration for not using this technique.

Thus it can be seen that no one technique fits all investigations. Where a crack is visible on the exposed surface it may just be sufficient to use surface breaking techniques, then repair the crack by its removal and making good by weld. On completion of the welding operation the repaired area should then be tested for surface breaking defects (this test should not be made until at least 12 hours has elapsed from the completion of all welding operations for materials/welding techniques susceptible to hydrogen embrittlement type cracking to allow for any weld induced cracks to propagate). At that point a second technique of volumetric testing to confirm the integrity of the body of the weld would be needed.

The techniques listed are not a comprehensive list and are meant for guidance; other techniques may be more applicable and may be successfully employed.

It is normal practice to combine magnetic particle investigation for surface and near surface breaking defects with an ultrasonic survey to ensure that a complete area survey is completed.

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When a Non Destructive Testing technique is to be used it is important to specify the extent of the tests to be carried out and the acceptance criteria to be employed and to ensure that the technicians employed to undertake the various required techniques are suitably qualified.

For guidance it is recommended that British Standard BS EN 12953 Design and Construction of Shell Boilers (the successor to BS 2790) is consulted. Whilst this document is designed to be used in the construction of new boilers it gives good guidance to the acceptable standards required for repairs. Whilst BS EN 12953 is a good source for guidance, other references may need to be used.

All techniques and acceptance levels must be agreed with the Competent Person, preferably before any tests are made.

Dye Penetrant Inspection

A low cost simple technique for detecting surface breaking defects such as cracks, weld overlap and porosity. It is a one-stop simple system, which relies on the skill of the technician.

The main drawback to this technique is the recording of results since the indications are continuously assessed from the application of the developer up a maximum period of around 15 minutes of it being applied. This limitation is to ensure that the results obtained are sharp and do not become confused with background leakage of dye.

Method of Application

- Surface cleaned (degreased mechanically cleaned of loose debris etc);
- Apply Penetrant;
- Remove excess penetrant;
- Apply Developer;
- Inspect test area; then,
- Clean area.

There are three major types of penetrant which should be applied to steel and copper for the following times prior to developing the results.

Material	Flaw	Water Washable Penetration Time	Post Emulsified Penetrant Time	Solvent Removed Penetration Time
Steel	Welds – Lack of Fusion, Porosity	20 minutes	20 minutes	20 minutes
Steel	Cracks	30 minutes	20 minutes	20 minutes
Steel	Fatigue Cracks	Not Applicable	30 minutes	20 minutes
Copper	Welds and plate defects	30 minutes	10 minutes	15 minutes
Copper	Fatigue Cracks	Not Applicable	30 minutes	15 minutes

The above table gives times for the application of penetrant for interpretation in normal light. Another similar process known as Fluorescent Dye Penetrant may be used for the detection of flaws as narrow as 150 nano-meters. This is a specialist technique, which must only be undertaken in clean conditions with suitably qualified staff.

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Magnetic Particle inspection (MPI)

This method is cost effective with excellent results for the detection of surface and near surface breaking defects in materials that can be magnetised. This is the primary NDT technique for crack detection.

The specimen to be tested is normally sprayed with a white contrast paint then a permanent magnet or electro–magnet poles are placed on the area to be tested. This in turn produces a magnetic flux within the material. This magnetic flux will, if the material is sound, remain within the material. If however there is a surface or near-surface breaking flaw the magnetic field in this area will become distorted causing a flux leakage in this area. This flux leakage can then be displayed by covering the area with fine iron particles. These particles will then collect in the area of the flux leakage and will show up against the white background paint as a dark line or dark area.

This method requires the magnet to produce magnetic force lines at a large angle to the expected defect. It is therefore normal to apply the magnetic force from two angles normally 90º to one another. Whilst it is normal for the items to be magnetised in the way stated, other methods, such as inducing a magnetic field with a coil, may be used.

The particles of iron used for these tests are normally applied in a liquid form (magnetic ink) but may be applied dry and are normally applied when the magnetizing method is active. The iron particles used for these tests should conform to BSEN 9934 Part2 in Specification for magnetic flaw detection inks and powders. This standard states that the particle size for 99% of a representative sample shall not exceed 100 microns and where in an ink form should be no less than 1·25% and no more than 3·5% by volume in concentration.

Magnets used for these tests should have the following characteristics:

- Permanent Magnets Lifting power 18Kg minimum with a pole spacing of 75 to 100 mm and a pull off force as applied to one pole equivalent to at least 9Kg.
- Electromagnets should possess a lifting power of 4·5 Kg for a pole spacing of 300 mm or less, with a pull off force of no less than 2·5 Kg for the same pole spacing.

Ultrasonic Techniques

Ultrasonic flaw detection, unlike the Dye Penetrant or MPI techniques, not only locates surfaces breaking defects but assesses the whole of the weld and parent material in the area under examination taking into account the volume of this area. This type of examination is often referred to as one of the volumetric techniques.

The ultrasonic technique uses a beam of short wavelength high frequency pulses in the frequency range of 1 to 5 MHz with wavelengths in the range 0·1 to 1 mm (typical values). The velocities of these waves through material depends very much on the type of material being tested but are normally ranged between 1000-6000 m/s. Such waves can travel large distances in fine grain metal.

In a normal simple detection technique a pulsed beam of ultrasound is sent through an item via a single hand held probe which is placed on the material's surface. A display screen then shows the time it takes for a pulse to travel to a reflector, which may be the back surface of the material, a defect or any other free surface. The received signal of a reflector may then be interpreted. The height of the reflected pulse relates to the flaw size, but its position and what type of defect is being seen will require considerable operator skill to interpret.

The sound waves so introduced into the material are normally of two types - compressional normally angled at 90º to the surface (0º probe), and shear (traverse). This will ensure complete scan coverage of the test area. The probes for these tests can be made such that the ultrasonic waves can be introduced to the material at nearly any angle; normally 45º, 60º and 70º probes are used (shear wave).

The main drawback to this technique is the fact that, in older instruments, a permanent record is not produced and the results rely on the skill of the operator. However modern Ultrasonic equipment is now able to produce a permanent record displayed in a more user friendly way and can record all actions and values taken by the operator which can be attached to the report of examination.

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It should be noted that readings obtained between the surface and around 2mm depth into the material should not be treated as accurate as figures obtained at depth greater than 2mm. Flaw sizing is generally recognised as better than ±2 mm.

Advantages

- High Sensitivity – can detect small flaws.
- Good penetration powers – can inspect thick sections.
- Accurate data – thickness measurements, location and defect size.
- Access – portable equipment.
- Inspection – test can be made from one surface only.

Disadvantages

- Rough surfaces – can limit the inspection.
- Geometry of Components – may present scanning inspection difficulties.
- Coarse Grain Structure – may present inspection (interpretation) difficulties particularly with castings and austenitic stainless steels etc.
- Results are interpreter/operator based and rely on the expertise and experience of the operator.

Radiography

Radiography uses X-ray or gamma rays to penetrate a material to produce a permanent image on a photographic film. The image so produced is the natural size of the object. This type of examination is often referred to as one of the volumetric techniques.

X-rays and gamma rays are very short wavelength rays of electromagnetic radiation. The rays produced have the ability to pass through solid material with some of the energy being absorbed by the material. These rays affect all types of photographic film.

If a photographic plate is placed on one side of the material and a source of electromagnetic radiation on the other side an image will be formed. The image formed will mirror the way in which the electromagnetic energy is absorbed by the material such that variations in thickness will show on the film with no differentiation as to whether these variations are in the surface or internal (not surface breaking).

The sources of radiation come basically in two forms:

> X-rays - are produced when an electric current is passed across a cathode-ray tube. The higher the voltage the greater the penetrating power of the radiation. It is common for industrial X-ray equipment to range in voltage from 20 KV to 20 MV. The most powerful X-ray equipment can be used to penetrate 20” (500 mm) of steel and produce satisfactory photographic images.
>
> Radioactive isotope Iridium-192 gamma-radiography is commonly used for most general engineering applications, but where greater penetrating powers are required Cobalt-60 is often used. Other isotopes are suitable for use with this technique.

To obtain the best image definition it is desirable to have a small diameter radiation source. Typical X-ray and gamma ray source sizes range from 1 to 4 mm which give good results. This technique is relatively expensive both for the equipment used and cost of the photographic film. It is worth noting that it is becoming more common for the image to be formed digitally doing away with the expense of film.

Generally gamma radiography requires a much longer exposure time than X-ray and produces a less sharp image. Gamma radiography however is less expensive and far more portable and is commonly used for site work.

Advantages:

- Provides a permanent record.
- Reveals internal flaws within materials without affecting the material.

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Disadvantages:

Normally access is required to both sides of the area to be tested.

Will not detect some flaws such as laminations.

Equipment is expensive.

Safety issues with the associated radiation.

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